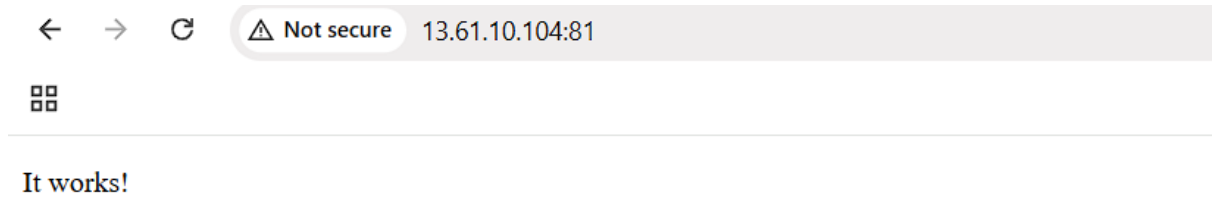


App and Web servers

1. Install nginx and run nginx on port number 81.



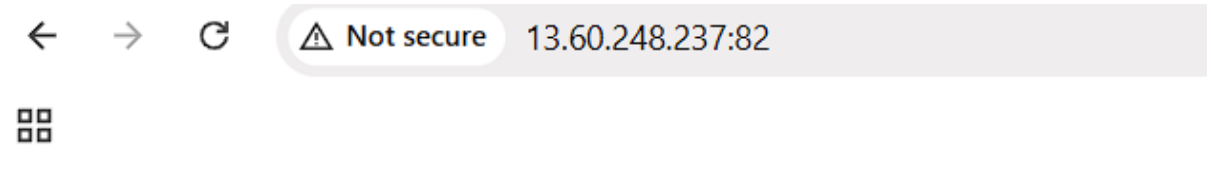
2. Deploy a sample index.html file on nginx.



3. Install Apache and run Apache on port number 82

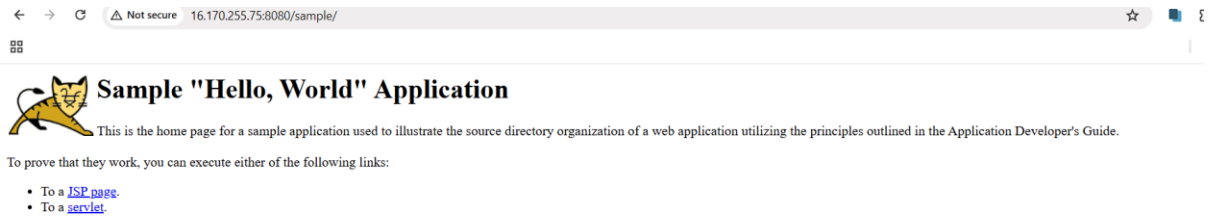


4. Deploy a sample index.html file on Apache..



5. Install Apache tomcat on port number 8082.

6. Deploy a sample app on webapps.



7. Create a tomcat.service file for tomcat..

```
[root@ip-172-31-43-188 tmp]# systemctl daemon-reload
[root@ip-172-31-43-188 tmp]# systemctl start tomcat
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
[root@ip-172-31-43-188 tmp]# readlink -f $(which java)
/usr/lib/jvm/java-17-amazon-corretto.x86_64/bin/java
[root@ip-172-31-43-188 tmp]# vi /etc/systemd/system/tomcat.service
[root@ip-172-31-43-188 tmp]# systemctl daemon-reload
[root@ip-172-31-43-188 tmp]# systemctl start tomcat
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
[root@ip-172-31-43-188 tmp]# systemctl status tomcat --no-paper -l
systemctl: unrecognized option --no-paper
[root@ip-172-31-43-188 tmp]# systemctl status tomcat -l
systemctl: invalid option -- 'l'
[root@ip-172-31-43-188 tmp]# systemctl status tomcat
* tomcat.service - Apache Tomcat 9
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
   Active: failed (Result: exit-code) since Fri 2026-09-04 15:02:34 UTC; 2min 14s ago
   Process: 26721 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=203/EXEC)
   CPU: 1ms
Sep 04 15:02:34 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: tomcat.service: Scheduled restart job, restart counter is at 5.
Sep 04 15:02:34 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: Stopped tomcat.service - Apache Tomcat 9.
Sep 04 15:02:34 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: tomcat.service: Start request repeated too quickly.
Sep 04 15:02:34 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: tomcat.service: Failed with result 'exit-code'.
Sep 04 15:02:34 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: Failed to start tomcat.service - Apache Tomcat 9.
[root@ip-172-31-43-188 tmp]# ls -l /opt/tomcat/bin/startup.sh
ls: cannot access '/opt/tomcat/bin/startup.sh': No such file or directory
[root@ip-172-31-43-188 tmp]# ls -l /opt/tomcat
total 16
drwxr-xr-x. 9 tomcat tomcat 16384 Sep  4 14:51 apache-tomcat-9.0.121
[root@ip-172-31-43-188 tmp]# mv /opt/tomcat/apache-tomcat-9.0.121/* /opt/tomcat/
[root@ip-172-31-43-188 tmp]# mv /opt/tomcat/apache-tomcat-9.0.121/.[]* /opt/tomcat/ 2>/dev/null
[root@ip-172-31-43-188 tmp]# rmdir /opt/tomcat/apache-tomcat-9.0.121
[root@ip-172-31-43-188 tmp]# chown -R tomcat:tomcat /opt/tomcat
[root@ip-172-31-43-188 tmp]# ls -l /opt/tomcat/bin/startup.sh
-rwxr-x--. 1 tomcat tomcat 1904 Aug 12 19:38 /opt/tomcat/bin/startup.sh
[root@ip-172-31-43-188 tmp]# systemctl daemon-reload
[root@ip-172-31-43-188 tmp]# systemctl start tomcat
[root@ip-172-31-43-188 tmp]# systemctl status tomcat
* tomcat.service - Apache Tomcat 9
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-09-04 15:09:46 UTC; 9s ago
   Process: 26995 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 27002 (java)
   Tasks: 30 (limit: 1059)
   Memory: 75.2M
   CPU: 4.147s
   CGroup: /system.slice/tomcat.service
           └─27002 /usr/lib/jvm/java-17-amazon-corretto.x86_64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager
Sep 04 15:09:46 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat 9...
Sep 04 15:09:46 ip-172-31-43-188.eu-north-1.compute.internal startup.sh[26995]: Tomcat started.
Sep 04 15:09:46 ip-172-31-43-188.eu-north-1.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat 9.
lines 1-14/14 (END)
```

← → ↻ ⚠ Not secure 16.170.255.75:8082 ☆ ⋮

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/11.0.25

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

- Tomcat Setup
- Realms & AAA
- Examples
- Servlet Specifications
- First Web Application
- JDBC DataSources
- Tomcat Versions

Managing Tomcat

For security, access to the [manager.webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 11.0 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 11.0 Documentation](#)
[Tomcat 11.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 11.0 Bug Database](#)
- [Tomcat 11.0 JavaDocs](#)
- [Tomcat 11.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

8. Configure HA Proxy server. ...

```
[root@ip-172-31-44-12 ~]# sudo haproxy -c -f /etc/haproxy/haproxy.cfg
[root@ip-172-31-44-12 ~]# sudo systemctl restart haproxy
[root@ip-172-31-44-12 ~]# sudo systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-09-04 07:57:58 UTC; 8s ago
   Process: 30082 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, st
 Main PID: 30118 (haproxy)
   Status: "Ready."
    Tasks: 3 (limit: 1059)
   Memory: 10.1M
      CPU: 137ms
   CGroup: /system.slice/haproxy.service
           └─30118 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/h
             30142 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/h

Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30118]: [ALERT] (30118) : config>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30118]: [NOTICE] (30118) : New wo>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30118]: [NOTICE] (30118) : Loadin>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal systemd[1]: Started haproxy.service - HAPro>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [ALERT] (30142) : backen>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
Sep 04 07:57:58 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
Sep 04 07:57:59 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
Sep 04 07:57:59 ip-172-31-44-12.eu-north-1.compute.internal haproxy[30142]: [WARNING] (30142) : Server>
[root@ip-172-31-44-12 ~]#
```